Arghamitra Talukder

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Education

2023 - 2028	Ph.D. in Computer Science, **Columbia University**	New York, NY
	Advisor: Itsik Pe'er, David A. Knowles	
	M.S. (2025) — *NSF CSGrad4US Fellow*	GPA: 3.98/4.0
2018 - 2021	B.S. in Electrical Engineering, **Texas A&M University**	College Station, TX
	Engineering Honors, Magna Cum Laude	GPA: 3.78/4.0

Projects

Learning Interpretable Representations via Privileged-Basis Embeddings

Designed a training framework that enforces hierarchical information allocation across embedding dimensions, yielding a privileged basis with ordered feature importance and improved interpretability of latent representations.

Predicting Exon Inclusion Levels via Contrastive Sequence Models

Developed a deep learning model to analyze gene regulation with a contrastive learning framework. This model identifies conserved functional elements by mapping evolutionarily related DNA sequences into a shared latent space, providing novel insights into gene expression.

Bayesian hierarchical isoform quantification with hybrid sequencing

Developed JOLI, a novel hierarchical model that enhances RNA transcript quantification by jointly integrating short-read (SR) and long-read (LR) sequencing data. The model utilizes a multi-sample learning framework to improve estimation accuracy and reproducibility, particularly for low-abundance isoforms.

Modeling cell plasticity with integrated single-cell data

Designed a statistical model to quantify cellular plasticity by integrating single-cell lineage and gene expression data. This approach utilizes a continuous annotation distribution methodology to statistically detect and score plastic cell states.

Publications

Arghamitra Talukder, Nicholas Keung, et al. (2026). "CLADES — Contrastive Learning Augmented Differential Splicing with Orthologous Positive Pairs". In: *RECOMB (Accepted)*. Online version.

Arghamitra Talukder, Shree Thavarekere, et al. (2025). "Multi-sample, multi-platform isoform quantification using empirical Bayes". In: *RECOMB-Seq (Short-talk)*. Online version.

Arghamitra Talukder and Itsik Pe'er (2024). "Computational Method to Detect Cancerous Plasticity". In: *IICD Workshop on Single-Cell Data Integration & OT, CRA-WP Grad Cohort for IDEALS (Short-talk)*.

Arghamitra Talukder, Rujie Yin, et al. (2022). "Does inter-protein contact prediction benefit from multi-modal data and auxiliary tasks?" In: *MLSB, NeurIPS*. Online version.

Ibrahim, Bassem, **Arghamitra Talukder**, and Roozbeh Jafari (2020). "Multi-source multi-frequency bio-impedance measurement method for localized pulse wave monitoring". In: *EMBC, IEEE*. Online version.

Work Experience

Computational Engineering Intern, LLNL June 2026 - Aug 2026

- Mentored by Alan Kaplan at *Lawrence Livermore National Laboratory (LLNL)*: Will develop machine learning methods for analysis of ALS-related datasets and evaluate model performance using statistical metrics.
- Will collaborate with interdisciplinary teams to support research planning, data analysis, and scientific discussions.

Product & Test Engineer, Texas Instruments June 2021 - Aug 2023

- Responsible for TI customed Ultra-High Voltage test technology with 70% cost efficiency.
- Created a product specific test program converting from legacy tester to the latest technology reducing 40% of the test time.
- Enabled traceability with the accuracy of 99.9% by device ID recovery obtained using OCR technology during internship.

Fellowships

RECOMB Travel Fellowship	May 2026
NIH T15 Training Grant	Sep 2025 - Aug 2027
CISE CSGrad4US Fellowship, CRA, NSF	Sep 2023 – Aug 2025 (Tenure) Sep 2027 – Aug 2028 (Tenure)

Teaching Assistant

Machine Learning for Functional Genomics	Sep 2025 - Dec 2025
Computational Genomics	Jan 2025 - May 2025
Experimental Physics Lab II-Mechanics	Jan 2019 - Dec 2019
Foundation of Engineering in Python and LabVIEW	Sep 2018 - Dec 2018

Skills

Programming & ML	Python, PyTorch, PyTorch Lightning, MATLAB, Perl, C, Bash
Genomics & Data	Genomic sequence processing, RNA-seq (short- and long-read)
Systems & Cloud	High-performance computing (SLURM, multi-GPU clusters), distributed training, Linux environments
Engineering & Testing	Python/Perl/C-based semiconductor test program development, legacy tester migration and test time reduction
Tools & DevOps	Git, Hydra/OmegaConf, WandB, Docker (basic)

Volunteering

Symposium volunteer at IICD Next-Generation Cancer Research	March, 2026
Columbia pre-submission application reviewer	Nov 2025, 2024
Conference volunteer at MLCB	Sep 2025
Panelist at the critical thinking in STEM career panel, NYGC	June 2024

Mentoring

Nicholas Keung, MS, Columbia CS	Jan 2025 - Ongoing
Abhinav Goel, Freshman, Columbia CS	Sep 2025 - Ongoing
Tanya Mehta, MS, Columbia CS	Sep 2025 - Dec 2025
Shree Thavarekere, MS, Columbia CS	Jan 2024 - Jan 2025
Claire Dawson, MEng, Columbia CS	Jan 2024 - May 2024